



System Requirements Specification (SRS)

for

Annual Analytical Quality Control (AQC) Portal

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1. Introduction

The **AQC Portal** manages the **Annual Analytical Quality Control (AQC) Exercise**, supporting different user roles such as **Admin, Organizing Divisions, Member Secretary, Chairman Control Board**, and **Participating Labs**. The portal ensures that only **EPA-recognized labs** participate in the AQC exercise through integration with the **EPA Recognition Portal**.

2. Purpose

This document outlines the system requirements for the **AQC Portal**, which will:

- Integrate with the **EPA Recognition Portal** to verify that only **EPA-recognized labs** participate in the AQC exercise.
 - Provide functionalities for the **Admin, Organizing Divisions, Member Secretary, Chairman Control Board**, and **Participating Labs**.
 - Manage the AQC process flow, including tasks like lab participation, sample dispatch, results submission, and result publication.
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3. System Overview

The **AQC Portal** is the central platform for managing the **AQC Exercise**. The portal will support the following features:

- **EPA Recognition Integration:** Ensures only EPA-recognized labs can participate.
 - **Role-Based Access Control:** Differentiated functionalities for each user role (Admin, Organizing Divisions, Member Secretary, Chairman Control Board, Participating Labs).
 - **Task Management:** Manages deadlines for participation, fee payment, sample dispatch, result submission, and result publication.
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4. User Roles and Permissions

4.1 Admin

- Full system access, including setting up tasks, assigning divisions, and managing deadlines.
- Assign organizing divisions for each AQC exercise.
- Integrate with the EPA Recognition Portal to verify lab participation.
- Manually add specific EPA-recognized labs if needed.

4.2 Organizing Divisions

- Manage the exercise, set dates, and oversee the dispatch of samples.
- Track lab participation and ensure compliance with deadlines.
- Publish results after validation.

4.3 Member Secretary

- Validate results submitted by labs, ensure they meet the required standards.
- Provide feedback to labs on their results and performance.
- Track submissions and ensure all labs meet deadlines.

4.4 Chairman Control Board

- Oversee the aggregation of data and ensure the results are accurate.
- Generate final reports, ensuring only EPA-recognized labs are included in the outcome.

4.5 Participating Labs

- Register for the AQC exercise through the portal.
- Submit their willingness to participate and pay fees.
- Receive sample kits from the organizing division, but submit their own results via the portal.
- Receive feedback and final reports.

5. Functional Requirements

5.1 EPA Recognition Integration

- The portal will integrate with the **EPA Recognition Portal** to verify lab recognition before allowing participation.
- If the lab is recognized, the system will proceed with registration; otherwise, an error message will be displayed, and the lab will not be allowed to participate.

5.2 Process Flow of the AQC Exercise

Step 1: Alerting Recognized Labs for Upcoming Exercise

- **Trigger:** The portal sends **email notifications** to all EPA-recognized labs regarding the upcoming AQC exercise.
- **Portal Notification:** A notice is posted on the portal for the labs, highlighting key dates and steps involved in the process.

Step 2: Lab Submission of Willingness to Participate

- **Action by Labs:** Participating labs submit their willingness to participate via the portal.
- **Submission to Organizing Divisions:** The labs' willingness to participate is displayed on the **Organizing Division's dashboard** for review.

Step 3: Payment of Participation Fees

- **Action by Labs:** Once the willingness is confirmed, labs submit the participation fee through the portal.
- **Payment Tracking:** The portal tracks payment completion and notifies the **Organizing Division** of the payment status.

Step 4: Sample Dispatch by Organizing Divisions

- **Action by Organizing Divisions:** Once fees are received, the **Organizing Division** sends out samples to participating labs.
- **Dispatch Date Capture:** The portal records the **dispatch date** for each lab when the samples are sent.

Step 5: Lab Submission of Sample Results

- **Action by Labs:** Labs receive the sample kits and perform their tests on the samples.
- **Submission of Results:** Labs then submit their **test results** for the received samples through the portal by the specified cutoff date.

Step 6: Result Publication by Organizing Divisions

- **Action by Organizing Divisions:** After the cutoff time for result submission, the **Organizing Division** validates and publishes the results.
- **Notification to Labs:** Labs are notified through the portal about the publication of the results.
- **Parameters Shared:** The published results include not only the lab's performance but also the parameters used for comparison.

Step 7: Feedback and Reports

- **Action by Member Secretary:** The **Member Secretary** reviews lab submissions and provides feedback on the results.
- **Final Report:** The **Chairman Control Board** reviews the exercise's aggregated data and publishes the final report to all participating labs.

6. Task Dates and Timelines

For each AQC exercise, the system will manage the following task dates:

1. **Willingness to Participate:**

- Start Date: [Admin defined]
 - End Date: [Admin defined]
 - 2. **Fee Payment:**
 - Start Date: [Admin defined]
 - End Date: [Admin defined]
 - 3. **Sample Dispatch:**
 - Dispatch Date: [Organizing Division defined]
 - 4. **Submission of Sample Results:**
 - Start Date: [Admin defined]
 - Cutoff Date: [Admin defined]
 - 5. **Result Publication:**
 - Start Date: [Admin defined]
 - End Date: [Admin defined]
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7. System Integration with EPA Recognition Portal

The **EPA Recognition Portal** will expose an **API** to allow the AQC Portal to verify if a lab is recognized.

- **API Integration:**
 - When a lab registers, the portal queries the **EPA Recognition API** to verify if the lab is EPA-recognized.
 - If recognized, registration proceeds; otherwise, the lab is not allowed to participate.
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8. Conclusion

This SRS outlines the requirements for the AQC Portal, incorporating the role-specific duties of the **Chairman Control Board** and **Member Secretary**. The portal integrates with the **EPA Recognition Portal** to ensure that only **EPA-recognized labs** participate in the **AQC Exercise**, while efficiently managing task deadlines, lab participation, sample dispatch, result submission, and the validation and publication of results.

AQC Portal Database Schema

1. Users Table

This table will store all users who access the portal (Admin, Organizing Divisions, Member Secretary, Chairman Control Board, Participating Labs).

Field Name	Type	Description
user_id	INT (PK)	Unique identifier for the user.
username	VARCHAR(255)	Username for login.
password_hash	VARCHAR(255)	Hashed password for secure authentication.
role	ENUM('Admin', 'Organizing Division', 'Member Secretary', 'Chairman Control Board', 'Participating Lab')	Role of the user in the portal.
email	VARCHAR(255)	User's email address.
first_name	VARCHAR(255)	User's first name.
last_name	VARCHAR(255)	User's last name.
status	ENUM('Active', 'Inactive')	User's status, whether they are active or inactive.

2. Labs Table

This table stores information about the labs, including their EPA recognition status.

Field Name	Type	Description
lab_id	INT (PK)	Unique identifier for the lab.
lab_name	VARCHAR(255)	Name of the lab.
epa_recognition_status	ENUM('EPA Recognized', 'Laboratories of Govt/SemiGovt/Public sector undertaking/Autonomous Bodies.')	Indicates if the lab is EPA recognized.

address	VARCHAR(255)	Address of the lab.
contact_number	VARCHAR(20)	Contact number of the lab.
email	VARCHAR(255)	Email address of the lab.
status	ENUM('Active', 'Inactive')	Indicates if the lab is currently active for participation.

3. AQC_Exercises Table

This table tracks details about each AQC exercise organized every year.

Field Name	Type	Description
exercise_id	INT (PK)	Unique identifier for the AQC exercise.
year	INT	The year the AQC exercise is conducted.
start_date	DATE	Start date for lab participation.
end_date	DATE	End date for lab participation.
status	ENUM('Upcoming', 'Ongoing', 'Completed')	Status of the exercise.

4. Participation Table

This table links labs to the AQC exercises they have participated in.

Field Name	Type	Description
participation_id	INT (PK)	Unique identifier for lab participation.
exercise_id	INT (FK)	References AQC_Exercises.exercise_id.

lab_id	INT (FK)	References Labs.lab_id.
willingness_status	ENUM('Confirmed', 'Pending', 'Rejected')	Lab's willingness to participate.
fee_paid	BOOLEAN	Whether the participation fee was paid.
sample_dispatched_date	DATE	Date the sample was dispatched to the lab.

5. Payments Table

This table tracks the payments made by the participating labs.

Field Name	Type	Description
payment_id	INT (PK)	Unique identifier for the payment.
participation_id	INT (FK)	References Participation.participation_id.
payment_date	DATE	Date the payment was made.
amount	DECIMAL(10, 2)	Amount paid by the lab.
payment_status	ENUM('Pending', 'Completed', 'Failed')	Status of the payment.

6. Samples Table

This table stores information about the sample kits sent to the participating labs.

Field Name	Type	Description
sample_id	INT (PK)	Unique identifier for the sample.
exercise_id	INT (FK)	References AQC_Exercises.exercise_id.

lab_id	INT (FK)	References Labs.lab_id.
sample_dispatch_date	DATE	Date the sample was dispatched.
sample_status	ENUM('Dispatched', 'Received', 'Tested')	Current status of the sample.

7. Results Table

This table stores the results submitted by the participating labs.

Field Name	Type	Description
result_id	INT (PK)	Unique identifier for the result entry.
sample_id	INT (FK)	References Samples.sample_id.
lab_id	INT (FK)	References Labs.lab_id.
test_result	TEXT	Test result submitted by the lab.
submission_date	DATE	Date the result was submitted.
status	ENUM('Pending', 'Validated', 'Rejected')	Status of the result.

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